INCH-POUND

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4 November 1996

PERFORMANCE SPECIFICATION

BURNER UNIT, FIELD FEEDING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a burner unit (stove) for use as the heat source for military field feeding equipment and field sanitation equipment (see 6.1).

2. APPLICABLE DOCUMENTS (None)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection as specified in 4.2.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, ATTN: SSCNC-WEF, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD FORM 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7310

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3.2 Operating requirements. Burner units (stoves) shall satisfy the following functional, operational and performance requirements.

3.2.1 Heat output. The stove shall have an infinitely adjustable firing rate. The minimum firing rate shall be not greater than 1.2 lbs-fuel/hour, and the maximum firing rate shall be not less than 2.8 lbs-fuel/hour. Heat output drift shall be not greater than ±10% during 3 hours of operation at any heat output setting.

3.2.2 Heat transfer. The stove shall be capable of increasing the temperature of 80 lbs of water 100°F in less than 35 minutes.

3.2.3 Compatibility. The stove shall be compatible with current Army field feeding equipment used in the Mobile Kitchen Trailer (MKT), NSN 7360-01-155-6020 and the Kitchen, Company Level Field Feeding (KCLFF), NSN 7360-01-200-9828, Modular Field Kitchen (MFK), NSN 7360-01-276-9817, and the Field Sanitation Center (FSC), NSN 7360-01-277-2558. The stove shall operate reliably in the burner rack (NSN 7360-01-250-3649) and base rack (NSN 7360-01-250-3652) with various appliances mounted, heater tank assembly (NSN 7360-01-248-6041), and model M59 field range outfit (NSN 7360-00-028-2153) in both upper and lower positions. Use of the stove shall require no modifications to field feeding equipment with the exception of an access hole for the electrical receptacles (see figures 1 through 8 for reference purposes only).

3.2.4 Fuel compatibility. The stove shall be capable of operating on JP-8.

3.2.5 Fuel capacity. The stove shall be capable of operating continuously, using only its internal fuel tank, at the highest heat setting, without refueling or shutting down, for not less than four hours.

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3.2.6 Weight. The total weight of the fully fueled stove shall be not greater than 68 pounds.

3.2.7 Grasping handle. A non-slip handle or grasp surface (e.g. grooved, knurled, or frictional) shall be located on the front of the stove to aid in removal from all cooking appliances.

3.2.8 Carrying handles. The stove shall be equipped with handles or grasp areas to allow two persons to lift and carry the unit. Handle length shall be a minimum of 11 inches to allow each person to use both hands to carry the stove.

3.2.9 Voltage. The stove shall operate only within the range of 22.0 to 29.5 volts DC.

3.2.10 Power consumption. The stove power consumption shall be minimized to allow operation from a rechargeable battery pack. Cumulative power consumption shall not be greater than 275 watt·hours to perform a cooking mission that includes refueling from an empty tank and a 3 hour cooking cycle at a firing rate that consumes the maximum amount of power. Maximum current draw under all circumstances shall not exceed 5 amps.

3.2.11 Reversed polarity. The stove shall be designed for protection against a reversed polarity power supply.

3.2.12 Burner control. The stove shall be equipped with a burner control that shall be used to initiate and stop combustion.

3.2.13 Refuel control. The stove shall be equipped with a refuel control that initiates the automatic refueling process. For safety purposes, the refuel control shall be designed such that the stove cannot refuel while the burner is operational.

3.2.14 Heat output control. The stove shall be equipped with a heat output control. The control shall vary the heat output linearly from minimum to maximum. The control shall have positive stops at the minimum and maximum settings. The relative heat output at different control settings shall be distinctly

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labeled and visually obvious. As a minimum, Low, Medium, and High settings shall be labeled. When the power control is moved to the "on" position, the stove shall automatically ignite, warm up, and operate without the need for additional operator input. The stove shall meet the heat output requirements (see 3.2.1) within two minutes. When the power control is moved from the "on" position to the "off" position, the stove shall automatically and immediately, (within 1 second), extinguish the flame and shut down all burner components. If the stove is equipped with a post-purge, the flame shall immediately extinguish, (within 1 second), and the post-purge must be complete within three minutes without damage to the stove or hazard to personnel.

3.2.15 Visual indicators. The stove shall be equipped with visual indicators to provide information about the operational status. The visual indicators shall illuminate as necessary to provide all visual signals specified in 3.2.16 thru 3.2.18. All visual indicators shall be clearly labeled for signal identification. All visual signals shall be visible from the front of the stove and shall be visible in all operational and environmental conditions specified herein. A means shall be provided to check all visual indicators prior to operation of the stove for indicator malfunction. Fuses/bulbs shall be readily accessible for easy removal and replacement without the use of special tools. All visual signals shall be colored as follows: Green: Power on; Red: Automatic shutdown or fault condition, and Amber: All other conditions.

3.2.16 Fuel level indicator. The stove shall be equipped with a fuel level indicator located on the front of the stove. The indicator shall visually alert the operator when there is 1/8 and less of usable fuel remaining in the tank. As an alternate, the indicator shall visually alert the operator when there is 30 minutes of remaining operational time.

3.2.17 Refuel indicator. The control panel shall provide a visual signal to indicate the stove is refueling and when the automatic refueling procedure is complete (full fuel).

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3.2.18 Fault indicator. Upon entering a fault shutdown (see 3.5.1), the stove shall provide a visual signal to alert the operator of the failure. The stove shall continue to identify the failure until the power control is returned to the "off" position.

3.2.19 Hour meter. The stove shall be equipped with an hour meter that accurately monitors the number of hours logged on the stove. The meter shall only operate when the power control is in the "on" position. The hour meter shall be readily visible while the stove is in all appliances.

3.2.20 Low fuel shutdown. The stove shall be designed such that it can be operated until a low fuel shut-down occurs without producing visible smoke and meeting the smoke requirements of 3.3.3. The stove shall cleanly and easily restart following the low fuel shutdown and subsequent refueling.

3.2.21 Automatic refueling. The stove shall be capable of filling its internal fuel tank through the fuel interface with a self priming integral fuel pump. Refueling from the low fuel shut down condition shall take not more that 5 minutes. Refueling shall automatically stop when the internal tank is full.

3.2.22 Manual refueling. The stove shall allow manual refueling using only the five-gallon fuel can and pouring spout (see 6.3).

3.2.23 Fuel draining. The stove shall be capable of being drained of fuel into a five-gallon fuel can (see 6.3) without the use of a funnel and without spilling in less than 5 minutes.

3.2.24 Conducted emissions, power leads, 10 kHz TO 10 MHz. The stove shall not produce conducted emissions in excess of the requirements for army ground equipment as specified in figure 9.

3.3 Safety requirements. The stove shall be safe to set up,operate, and transport and shall not create any personal hazards.

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3.3.1 Fault shutdown. To prevent buildup of combustible vapors of liquids, the stove shall automatically shut down whenever the burner flame is lost and cannot be reacquired safely.

3.3.2 Noise. The stove shall not produce noise in excess of 78 dBA when operated in the burner stand, base, and cradle with stock pot on a hard surface. Reflected noise shall be minimized.

3.3.3 Smoke. The Bacharach smoke number shall be not greater than 1 during the startup phase and shall not be greater than 0 at any other time.

3.3.4 Emissions. The stove shall not produce emissions, corrected to 3% O_2, in excess of 1 ppm HCOH, 40 ppm SO_2, 150 ppm CO, and 18 ppm NO_2.

3.3.5 Surface temperatures. Control surfaces shall be designed to prevent burns to operators.

3.3.6 Overload protection. Necessary overload protection (fuse or circuit breaker) shall be incorporated to protect against component damage, electrical short, and personnel burns. The overload protection shall be readily accessible for resetting or removal/replacement. If fuses are used, they shall be protected from accidental grounding from rings, watches, or tools. No other burner component shall be removed in order to gain access to fuses or circuit breakers. No special tools shall be required for fuse replacement. Spare fuses shall be located adjacent to the fuse holder.

3.3.7 Inadvertent ignition. The stove shall not initiate the ignition sequence if power is connected to the stove while the power control is in the "on" position. To initiate ignition after all fault shutdowns, the burner control shall be cycled through the "off" position.

3.3.8 Sharp edges/moving parts. Non-functional sharp edges and projections shall be eliminated. The stove shall provide protection to the operator from all moving parts.

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3.3.9 Health hazards. The stove shall be designed to minimize the danger of fuel leaks, flare ups, and hazards to operators. Damage to safeguards shall be easily noticed by operators. The stove shall be designed to withstand common spills without damage. The stove fuel system shall be designed to withstand grease spills and possible grease fires without loss of integrity.

3.4 Interface requirements. Each stove shall satisfy the following interface requirements.

3.4.1 Dimensions. The dimensions of the stove from the front view shall be between 18.5 and 19 inches width. The height shall not exceed 9.625 inches.

3.4.2 Location of controls and indicators. All controls shall be readily accessible and easily operable and all warning indicators shall be easily visible when the burner is installed in the top position of the model M59 field range outfit (NSN 7360-00-082-2153) with the range door closed and the sliding access panels open. The controls shall be designed and located so that they are not susceptible to being moved accidentally, particularly critical controls whose inadvertent operation might cause damage to equipment, injury to personnel or degradation of system functions. Controls and indicators shall not protrude beyond the framework of the stove to prevent damage.

3.4.3 Fuel interface. The stove fuel interface fitting shall be compatible with Parker coupling, part number BH2-60Y. The fuel interface shall be located on the front of the stove and shall be accessible when the stove is installed in all cooking appliances. The fuel interface shall not protrude beyond the framework of the stove. The fuel interface shall provide a positive shut off when disconnected from the external fuel line.

3.4.4 Labeling. Each control and indicator shall be clearly labeled using the English language to indicate its intended function and shall be located on or very near the item which it identifies. Labels shall be oriented horizontally so that they may be read quickly, accurately, and easily from left to right at

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the anticipated operational reading distances. Labeling shall not be obscured by other equipment when the stove is installed for its intended use. Labels shall be capable of withstanding fuels, cleaning products, and the cleaning process without degradation.

3.4.5 Instructions. The stove shall have permanently affixed, complete operating and refueling instructions and warning labels that are easily read by the operator when the stove is installed in all cooking appliances.

3.4.6 Power receptacles. The stove shall be equipped with two power receptacles. The input receptacle shall be compatible with plug, MS3106A16-11S. The output receptacle shall be compatible with plug, MS3106A16-11P. The output receptacle shall be capable of supplying sufficient power for a chain of 3 additional stoves. The receptacle shall be located on the rear not less than 1.5 inches from any edge of the envelope specified in 3.4.1 and not more that 5.5 inches from the right side of the rear view of the envelope. The receptacles shall be easily and readily accessible for connection of the compatible plugs when the unit is installed in all appliances.

3.4.7. Operator interface. The operator shall be capable of performing all actions necessary to carry, install, operate, and remove the stove while bare-handed and while wearing cold weather gloves, NSN 8415-01-310-7338.

3.5 Operating environmental conditions. Burner units (stoves) shall operate under the following environmental conditions.

3.5.1 Low temperature operation. The stove shall operate in ambient temperatures as low as -25°F.

3.5.2 High temperature operation. The stove shall operate in ambient temperatures as high as 120°F.

3.5.3 Altitude. The stove shall operate in altitudes up to 8,000 feet above sea level.

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3.5.4 Rain. The stove shall operate after being exposed to 4 in/hr rain.

3.5.5 Wind. The stove shall operate in wind up to ten miles per hour.

3.5.6 Vibration. The stove shall operate after exposure to vibrations characteristic of transportation as unrestrained cargo on ground vehicles.

3.5.7 Shock. The stove shall operate after being dropped five times from the top burner position of the M59 field range.

3.5.8 Storage. The stove shall be capable of withstanding storage temperatures as high as 160°F and as low as -50°F without failing to operate, causing permanent set or loss of resiliency of gaskets, or discoloring, cracking, or bulging of any component of the stove.

3.5.9 Non-level operation. The stove shall operate when tilted up to 4 degrees on each of its four sides.

3.6 Ownership and support requirements. The stove shall meet the following requirements.

3.6.1 Reliability. The stove shall be capable of demonstrating a mean time between essential function failure (MTBEFF) of at least 300 operational hours.

3.6.2 Maintainability. The stove shall be capable of demonstrating a maximum total maintenance ratio of 0.0063. The maintnance ratio shall be the total time required for unit and depot, scheduled and unscheduled maintenance actions divided by the total operational time. The operators manual shall include simple procedures to identify and replace any failed component. All user repairable/replaceable components shall be identified, accessed, and repaired/replaced withing 30 minutes of the failure with minimum disturbance to adjacent components in the stove and without usage of special tools.

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3.6.3 Cleaning. Procedures for cleaning the stove, shall be provided and detailed in the operators manual, using commonly available cleaning supplies.

3.6.4 Nameplate. The stove shall have a nameplate marked with the National Stock Number and the manufacturer's name, part number, and date or manufacture.

3.6.5 Preventive Maintenance Checks and Services (PMCS). Daily PMCS as described in the operators manual shall not exceed 10 minutes.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).
c. End item examination (see 4.4).

4.2 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined and tested as specified in column "FAT" of table I. Sampling for first article inspection shall be as specified in the contract or purchase order (see 6.2 and 6.6). Any nonconformance shall be cause for rejection of the first article.

4.3 Conformance inspection. Conformance inspection shall include the examinations and tests as specified in column "CT" of table I. Sampling for conformance inspection shall be as specified in the contract or purchase order (see 6.2 and 6.7).

4.4 Acceptance test. Acceptance examination and test shall be as specified in column "AT" of table I. Sampling for test and inspection shall be as specified in the contract or purchase order (see 6.2).

4.5 Verification alternatives. Alternative test methods, techniques, or equipment, including application of statistical

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process control, tool control, or cost effective sampling procedures may be proposed by the contractor. Acceptable alternative verification approach, including the use of sample testing shall be identified in the contract.

Table I. Requirements vs. verification methods.

Title	Requirement	Verification	FAT	CT	AT
OPERATING REQUIREMENTS					
Heat output	3.2.1	4.6.1	X	X	X
Heat transfer	3.2.2	4.6.2	X	X	
Compatibility	3.2.3	4.6.3	X	X	
Fuel compatibility	3.2.4	4.6.4	X	X	
Fuel capacity	3.2.5	4.6.5	X	X	
Weight	3.2.6	4.6.6	X	X	
Grasping handle	3.2.7	4.6.7	X	X	
Carrying handles	3.2.8	4.6.8	X	X	
Voltage	3.2.9	4.6.9	X	X	X
Power consumption	3.2.10	4.6.10	X	X	
Reversed polarity	3.2.11	4.6.11	X	X	
Burner control	3.2.12	4.6.12	X	X	
Refuel control	3.2.13	4.6.13	X	X	
Heat output control	3.2.14	4.6.14	X	X	
Visual indicators	3.2.15	4.6.15	X	X	
Fuel level indicator	3.2.16	4.6.16	X	X	
Refuel indicator	3.2.17	4.6.17	X	X	
Fault indicator	3.2.18	4.6.18	X	X	

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Title	Requirement	Verification	FAT	CT	AT
Hour meter	3.2.19	4.6.19	X	X	
Low fuel shutdown	3.2.20	4.6.20	X	X	X
Automatic refueling	3.2.21	4.6.21	X	X	X
Manual refueling	3.2.22	4.6.22	X	X	
Fuel draining	3.2.23	4.6.23	X	X	
Conducted emissions	3.2.24	4.6.24	X	X	
SAFETY REQUIREMENTS					
Fault shutdown	3.3.1	4.6.25	X	X	
Noise	3.3.2	4.6.26	X	X	
Smoke	3.3.3	4.6.27	X	X	
Emissions	3.3.4	4.6.28	X	X	
Surface temperatures	3.3.5	4.6.29	X	X	
Overload protection	3.3.6	4.6.30	X	X	
Inadvertent ignition	3.3.7	4.6.31	X	X	
Sharp edges/moving parts	3.3.8	4.6.32	X	X	X
Health hazards	3.3.9	4.6.33	X	X	
INTERFACE REQUIREMENTS					
Dimensions	3.4.1	4.6.34	X	X	
Location of controls	3.4.2	4.6.35	X	X	
Fuel interface	3.4.3	4.6.36	X	X	
Labeling	3.4.4	4.6.37	X	X	X
Instructions	3.4.5	4.6.38	X	X	

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Title	Requirement	Verification	FAT	CT	AT
Power receptacles	3.4.6	4.6.39	X	X	
Operator interface	3.4.7	4.6.40	X	X	
OPERATING ENVIRONMENTAL CONDITIONS					
Low temperature operation	3.5.1	4.6.41	X	X	
High temperature operation	3.5.2	4.6.42	X	X	
Altitude	3.5.3	4.6.43	X	X	
Rain	3.5.4	4.6.44	X	X	
Wind	3.5.5	4.6.45	X	X	
Vibration	3.5.6	4.6.46	X	X	
Shock	3.5.7	4.6.47	X	X	
Storage	3.5.8	4.6.48	X	X	
Non-level operation	3.5.9	4.6.49	X	X	
OWNERSHIP AND SUPPORT REQUIREMENTS					
Reliability	3.6.1	4.6.50	X	X	
Maintainability	3.6.2	4.6.51	X	X	
Cleaning	3.6.3	4.6.52	X	X	
Nameplate	3.6.4	4.6.53	X	X	X
PMCS	3.6.5	4.6.54	X	X	

4.6 Methods of inspection. For the purpose of testing, unless otherwise specified, the "standard test setup" (see figure 1) will refer to a stove installed in a standard setup consisting

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of the burner rack (NSN 7360-01-250-3649), base rack (NSN 7360-01-250-3652), closed cradle (NSN 7330-01-248-9964), 15 gallon stock pot (NSN 7330-00-292-2307) filled with 80 pounds of water, and cooking pot cover (NSN 7330-00-250-6300). Ambient temperature shall be between 60°F and 85°F. Wind conditions shall be still. The stove shall be connected to a 26.0 ± 2.0 volts DC power supply. Altitude shall be less than 1,000 feet above sea level. A "standard cycle" shall include: An initial condition of not more than half a tank of fuel; automatic refueling; startup; heating for 3 hours; 1 hour at each, maximum, medium, and minimum; shutdown; and refueling. The fuel used shall be JP-8.

4.6.1 Heat output test. The stove shall be set up on a scale with a full fuel tank. The stove shall be ignited (t=0) and operated at the lowest heat output control setting for three hours. The weight of the burner shall be recorded every five minutes, the first measurement being at t=5 minutes. The heat output shall be considered the average heat output during the duration of each test. This test shall be repeated at the highest heat output control setting. Verify that the stove fires as required.

4.6.2 Heat transfer test. The standard test setup as specified in 4.6 shall be used. A temperature probe shall be placed in the center of the stock pot, halfway between the bottom of the pot and the surface of the water. The initial water temperature shall be within 10°F of ambient. The stove shall be ignited (t=0). Failure to achieve at least a 100°F water temperature increase within 35 minutes will constitute a failure of this test.

4.6.3 Compatibility test. This test will be conducted in conjunction with the reliability test in 4.6.50. The stove will be operated in each piece of equipment listed in 3.2.3 for a standard cycle as specified in 4.6. Verify that the stove fits, and functions as required in all pieces of equipment.

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4.6.4 Fuel compatibility test. All tests shall be performed using JP-8 fuel. Verify that the stove is capaable of operating with JP-8 fuel.

4.6.5 Fuel capacity test. The standard test setup as specified in 4.6 shall be used. The stove shall be fully fueled using the automatic refueling system. The stove shall be ignited (t=0) and operated at the maximum heat output until a low fuel shutdown occurs. Verify that the stove can operate for at least four hours.

4.6.6 Weight verification. The stove shall be fully fueled using the automatic refueling system. Verify that the weight of a fully fueled stove is no greater than 68 pounds.

4.6.7 Grasping handle verification. Verify that a grasp handle is located on the front of the stove.

4.6.8 Carrying handles verification. Measure and verify the handle length dimensions, and that the stove can be carried easily by two persons.

4.6.9 Voltage test. The voltage shall be verified as follows.

4.6.9.1 Low voltage test. The stove shall be connected to a voltage supply set at 20.0 ± 1.0 volts. The refueling procedure shall be attempted. Failure to shut down immediately with a fault or low voltage indicator shall constitute a failure of this test. Burner ignition shall be atttempted. Failure to shut down immediately with a fault or low voltage indicator shall constitute a failure. All controls shall be reset to the "off" position. The voltage shall be set to 24 ± 1.0 volts. The burner shall be refueled and ignited. The voltage shall be decreased until the unit automatically shuts down due to low voltage. Verify that the stove shuts down when the voltage falls below 22 volts ± 0.5 volts.

4.6.9.2 Overvoltage test. The stove shall be connected to a voltage supply set at 31.0 ± 1.0 volts. The refueling procedure

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shall be attempted. Failure to immediately shut down with a fault or over voltage indicator shall constitute a failure of this test. All controls shall be reset to the "off" position. The voltage shall be set to 24 ± 1.0 volts. The burner shall be refueled and ignited. The voltage shall be increased until the unit automatically shuts down due to over voltage. Verify that the stove shuts down when the voltage goes above 29.5 ± 0.5 volts.

4.6.10 Power consumption test. The standard test setup as specified in 4.6 shall be used. The stove shall be operated through a standard cycle as specified in 4.6. Input current and voltage shall be monitored throughout the duration of the test. Verify that the stove operates within the power consumption rate and that the maximum current drawn does not exceed 5 amps.

4.6.11 Reversed polarity test. The stove shall be connected to a reversed polarity 24 volt d.c. power supply. The refuel cycle shall be attempted. Burner ignition shall be attempted. The stove shall then be connected to a 24 volt d.c. power supply with correct polarity. Burner refueling and ignition shall be performed. Failure to operate properly after being connected to the corrected power supply without operator corrective action shall constitute a failure of this test.

4.6.12 Burner control verification. Verify that the stove is equipped with a burner control.

4.6.13 Refuel control verification. Verify that the stove is equipped with a refuel control.

4.6.14 Heat output control verification. Verify conformance to section 3 in conjunction with the reliability test 4.6.50. Verify that the stove is equipped with a heat output control, and that it operates as specified.

4.6.15 Visual indicators verification. Verify that the stove is equipped with visual indicators to provide information about its operational status, and also that it provides a means to check for indicator malfunction.

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4.6.16 Fuel level indicator test. The standard test setup as specified in 4.6 shall be used. The stove shall be fully fueled using the automatic refueling system. The stove shall be operated at the maximum firing rate until the operator is visually alerted of a low fuel condition. The stove shall be operated at the same firing rate until flame loss due to low fuel. Verify that the stove is equipped with a fuel level indicator, and that it operates as specified.

4.6.17 Refuel indicator verification. Verify that the stove has a refuel indicator as required.

4.6.18 Fault indicator verification. Verify that the stove is equipped with a fault indicator providing a visual signal to alert the operator of failure.

4.6.19 Hour meter test. Verify that the stove is equipped with an hour meter and that it is readily visible after the stove is installed in all appliances. In addition, accuracy of the meter shall be monitored in conjunction with the reliability test, 4.6.50. Accuracy whall be within ± 5% of actual time.

4.6.20 Low fuel shutdown test. The standard test setup as specified in 4.6 shall be used. The stove shall be operated until low fuel shutdown occurs. The stove shall then be refueled and restarted. Verify that the stove cleanly and easily restarts (no visible smoke is produced) following a low fuel shutdown, refueling, and restart.

4.6.21 Automatic refueling. A stove completely empty of fuel shall be connected to a five gallon fuel can using the suction adapter assembly and external fuel line (see 6.3). The level of the bottom of the stove shall be 5 feet higher than the level of the bottom of the fuel can. The stove shall be switched to the "refuel" position. The stove shall be allowed to refuel until it automatically stops. Failure to self-prime and to automatically stop shall constitute a failure or this test. The fuel line shall be disconnected, and the stove shall be operated until low fuel shutdown occurs. The fuel line shall be reconnected, and the power control shall be moved to the "refuel" position (t=0).

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The time required to completely refuel shall be recorded. Verify that the stove is capable of automataic refueling, and that refueling automatically stops with the internal tank is full.

4.6.22 Manual refueling test. The stove shall be refueled using the fuel can and pouring spout (see 6.3). Verify that the stove can be manually refueled from a five-gallon fuel can without spilling or requiring the use of any tools in the allotted time.

4.6.23 Fuel draining test. The stove with a full fuel tank shall be drained as outlined in the operators manual. Verify that the tank can be completely drained in less than 5 minutes, including time to access drain, without the use of a funnel, and without spilling fuel.

4.6.24 Conducted emissions, power leads 10 kHz TO 10 MHZ test. This test is used to verify that electromagnetic emissions from the stove do not exceed the requirements specified in figure 9 for power input leads, including returns. Verify that the stove operates within the limits specified in figure 9.

4.6.24.1 Test equipment. The test equipment shall include measurement receiver, data recording device, signal generator, attenuator, 20 dB, oscilloscope, and Line Impedance Stabilization Network (LISN) (see figures 5 and 6).

4.6.24.2 Test setup. Maintain a basic test setup for the stove as shown(see figure 10), and as follows:

4.6.24.2.1 Stove configuration. The stove shall be configured as shown in the general test setup (see figure 10). The setup shall be maintained during all testing unless other direction is given in the contract for a particular test method.

4.6.24.2.2 Orientation of stove. The stove shall be located 4.00 ± 0.75 inches from the front edge of the ground plane subject to allowances for providing adequate room for cable arrangement as specified below.

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4.6.24.2.3 Construction and arrangement of stove cables. Electrical cable assemblies shall simulate actual installation and usage. Shielded cables or shielded leads (including power leads and wire grounds) within cables shall be used only if they have been specified in installation instructions. Cables shall be checked against installation requirements to verify proper construction techniques. Details on the cable construction used for testing shall be included in the test plan. Two meters of input power leads (including returns) shall be routed parallel to the front edge of the setup. Remaining cable lengths shall be routed to the back of the setup and shall be placed in a zigzagged arrangement. The power leads shall be connected to the LISNs. After the 6 feet exposed length, the power leads shall be terminated at the LISNs in as short a distance as possible. The total length of power lead from the stove electrical connector to the LISNs shall not exceed 8 feet. All power leads shall be supported 2 inches above the ground plane.

4.6.24.2.4 Electrical interface. The electrical output shall not be used for this test.

4.6.24.3 Operation of stove. During conducted emission measurements, the stove shall be operated in both the refueling mode and burner operating mode (at high, medium, and low).

4.6.24.4 Calibration. Configure the test setup for the measurement system check as shown in figure 11. Ensure that the stove power source is turned off. Connect the measurement receiver to the 20 dB attenuator on the signal output port of the LISN.

4.6.24.5 Stove testing. Configure the test setup for compliance testing of the stove as shown in figure 12. Connect the measurement receiver to the 20 dB attenuator on the signal output port of the LISN.

4.6.24.6 Test procedures. The test procedures shall be as follows:

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4.6.24.6.1 Calibration. Perform the measurement system check using the measurement system check setup of figure 11. Turn on the measurement equipment and allow a sufficient time for stabilization. Apply a calibrated signal level, which is 6 dB below the figure 9 limit at 10 kHz, 100 kHz, 2 MHZ and 10 MHZ to the power output terminal of the LISN. Also, verify that the voltage waveform is sinusoidal. Scan the measurement receiver for each frequency in the same manner as a normal data scan. Verify that the measurement receiver indicates a level within ±3 dB of the injected level. Correction factors shall be applied for the 20 dB attenuator and the voltage drop due to the LISN 0.25 microfarad coupling capacitor. If readings are obtained which deviate by more than ±3 dB, locate the source of the error and correct the deficiency prior to proceeding with the testing.

4.6.24.6.2 Stove testing. Perform conducted emission data scans using the measurement setup of figure 12. Turn on the stove and allow a sufficient time for stabilization. Select an appropriate lead for testing. Scan the measurement receiver over the applicable frequency range, using the bandwidths and minimum measurement times in table II. Repeat for each power lead.

Table II. Bandwidth and measurement time.

Frequency Range	6 dB Bandwidth	Dwell Time	Minimum Measurement Time Analog Measurement Receiver
10 kHz - 250 kHz	1 kHz	0.015 sec	0.015 sec/kHz
250 kHz - 10 MHz	10 kHz	0.015 sec	1.5 sec/kHz

4.6.24.6.3 Data presentation. Data presentation shall be as follows: Continuously and automatically plot amplitude versus frequency profiles on X-Y axis outputs. Manually gathered data is not acceptable except for plot verification. Display the applicable limit on each plot. Provide a minimum frequency resolution of 1% or twice the measurement receiver bandwidth, whichever is less stringent, and a minimum amplitude resolution of 1 dB for each plot. Provide plots for both the measurement system check and measurement portions of the procedure.

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4.6.25 Fault shutdown. The stove shall be operational with a stable flame at the start of this test. A fault condition shall be simulated by disabling a critical function (e.g., blocking the air supply or fuel supply, or disabling an essential component) of the stove, causing a flame loss. Stove behavior shall be observed. When the shutdown is obviously complete, the power control shall be moved to the "off" position. Any of the following shall constitute a failure of this test: failure to automatically enter fault shutdown after flame loss, failure to shut down all burner components within three minutes after entering fault shutdown, or failure to immediately shut down all components that may cause fuel to enter the combustion chamber. Any of the following shall constitute a failure of the fault indicator test (4.6.18): failure to immediately provide a visual signal identifying the failure, or failure of visual signal to continue identifying the failure until power control is returned to the "off" position.

4.6.26 Noise test. This test shall be conducted in conjunction with the realiability test in 4.6.50. The stove shall be operated for a standard test cycle as specified in 4.6. The burner unit may be set up in a location to eliminate all reflected noise with the exception of noise reflected off of the floor surface. The floor surface shall be concrete, asphalt, or wood, a minimum of 8 feet by 8 feet in surface area. The test stand shall be set up in the center of the floor surface. The noise level shall be measured 5 feet 3 inches above the floor surface, and at a distance of 2 feet in front of the burner stand. Verify that the noise level of the stove is not greater than 78 dBA.

4.6.27 Smoke test. This test shall be performed in conjunction with the low temperature operation test in 4.6.40. The stove shall be operated in the standard test setup as specified in 4.6 with the following exception. The combustion product collector (figure 15) shall be used in place of the 15 gallon stock pot. The stove shall be operated for a standard cycle as specified in 4.6. A Bacharach smoke number shall be taken within the first 30 seconds of ignition, again at 90

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seconds, then every 15 minutes for the duration of the cooking phase, and within 30 seconds after shutting off the burner. Verify that the stove's Bacharach smoke numbers meet the specified requirement.

4.6.28 Emissions test. The stove shall be operated in the standard setup as specified in section 4.6 with the following exception. The combustion product collector (figure 15) shall be in place of the 15 gallon stock pot. The stove shall be operated for a standard cycle as specified in 4.6. Ignition shall be initiated at t=0. Beginning at t=30 seconds and approximately every 15 minutes for the remainder of the test, measurements shall be taken for the following gases: CO, NO_2, SO_2, HCHO, and O_2. The time weighted average for the entire cooking cycle shall be computed. The measurement for CO, NO_2, SO_2, HCHO shall be corrected to 3% O_2 using the following formula.

$$X\%ppm = Y\%ppm\ [(21\%O_2 - 3\%O_2)/(21\%O_2 - Z\%O_2)]$$

Where:

X = ppm of the emission in question, corrected to 3% O_2
Y = ppm of emission as measured in exhaust stream
Z = concentration (as a percent) of O_2 in the exhaust stream

Ambient air is assumed to be 21% O_2

Verify that the emissions emitted are within the specified requirements.

4.6.29 Surface temperatures. Control surface temperatures shall be monitored during the reliability testing in 4.6.50. Exceeding the limits of 140°F for metal surfaces, momentary contact, 115°F for metal surfaces, prolonged contact, and 156°F for plastic will constitute a failure of this test.

4.6.30 Overload protection verification. Verify that the stove is equipped with fuses or circuit breakers, that spare fuses are available, and that no special tools are required for fuse replacement.

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4.6.31 Inadvertent ignition test. The stove shall be connected to a variable voltage power supply. The voltage shall be set at 25 ± 2 volts. The stove shall be operated for 5 minutes. The voltage shall be decreased until a low voltage shutdown occurs. The voltage shall then be increased until it reaches at least 25 ± 2 volts dc. Reignition or attempted reignition of the stove shall constitute a failure of this test. The power control shall then be moved from "on" to "off" and back to "on". Failure to reignite shall constitute a failure of this test. The stove shall be operated for 5 minutes. The voltage shall be increased until an over voltage shutdown occurs. The voltage shall then be decreased until it reaches 25 ± 2 volts dc. Reignition of the stove shall constitute a failure of this test. The power control shall then be moved from "on" to "off" and back to "on". Failure to reignite shall constitute a failure of this test.

4.6.32 Sharp edges/moving parts verification. Verify that the stove has no sharp edges or hazards, and that the operator is protected from contact with moving parts when the stove is on a flat surface.

4.6.33 Health hazards. This test will be conducted in conjunction with the reliability test in 4.6.50. The stove will be inspected for the presence of required safety devices and for absence of hazards, to include electrical, fuel, and appliance interfacing. Tests will be made for safety device operation, toxic gases, and component temperatures. All incidents occurring during testing will be analyzed for safety implications. Any "high risk" or "medium risk" safety or health hazard (as defined in 4.6.32.1) attributable to the stove shall constitute a failure of this test.

4.6.33.1 Risk assessment. Risk assessment shall be based on the combination of hazard severity and hazard probability as defined in 4.6.33.1.1 and 4.6.33.1.2. Risk level shall be IIE, IIID, IIIE, or IVB-IVE.

4.6.33.1.1. Hazard severity. Hazard severity categories are defined to provide a qualitative measure of the worst credible

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mishap resulting from personnel error; environmental conditions; design inadequacies; procedural deficiencies; or system, subsystem or component failure or malfunction as follows:

Category		
I	Catastrophic	May cause death or system loss.
II	Critical	May cause severe injury, severe occupational illness, or major system damage
III	Marginal	May cause minor injury, minor occupational illness, or minor system damage
IV	Negligible	Will not result in injury, occupational illness, or system damage

4.6.33.1.2 Hazard probability. The probability that a hazard will be created during the planned life expectancy of the system shall be described in potential occurrences. A qualitative hazard probability may be derived from research, analysis, and evaluation of historical safety data from similar systems. Hazard probability ranking shall be:

Description	Level	Individual Item
FREQUENT	A	Likely to occur frequently
PROBABLE	B	Will occur several times
OCCASIONAL	C	Likely to occur sometime
REMOTE	D	Unlikely but possible to occur in life of an item
IMPROBABLE	E	So unlikely, it can be assumed occurrence may not be experienced

4.6.34 Dimensions verification. Measure and verify the exterior dimensions of the stove.

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4.6.35 Location of controls verification. Verify conformance by placing the stove in the top position of the M59 range cabinet (see 6.3) closing the door and opening the access panels and operating the controls. Verify visibility of warning indicators. The stove shall then be stood on the front end on a flat surface. Verify that the stove controls and indicators do not protrude beyond the framework of the stove.

4.6.36 Fuel interface verification. Verify that the stove fuel interface fitting is compatible with the Parker coupling, and that it is located on the front of the stove.

4.6.37 Labeling verification. Verify that controls and indicators are clearly labeled in English, and located as indicated.

4.6.38 Instructions verification. This test may be conducted in conjunction with the reliability test in 4.6.50. Verify that the stove can be operated and refueled per the affixed instructions, and that they can be easily read by the operator in all cooking appliances.

4.6.39 Power receptacles verification. Verify that the stove's input and output receptacles are compatible with the plugs specified, and that they are in the proper location.

4.6.40 Operator interface verification. Verify that all controls, handles and other operator interfaces can be operated and accessed using gloved and bare hands.

4.6.41 Low temperature operation test. The standard test setup as specified in 4.6 shall be used. The entire assembly including the stove, fuel, and fuel lines shall be cold soaked for 4 hours at -25°F. The stove shall perform a standard operational cycle as specified in 4.6. The operation cycle shall be repeated for a total of two tests. When the smoke generation test (see 4.6.27) is performed in conjunction with this test, one cycle shall be performed with the smoke collection apparatus. Verify that the stove operates at an ambient temperature of -25°F.

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4.6.42 High temperature operation test. The standard test setup as specified in 4.6 shall be used. The entire assembly including the stove, fuel, and fuel lines shall be hot soaked for 4 hours at 120°F. The unit shall perform a standard cycle as specified in 4.6. The operation cycle shall be repeated for a total of two tests. Verify that the stove operates at an ambient temperature of 120°F.

4.6.43 Altitude test. The standard test setup as specified in 4.6 shall be used. The entire assembly including the stove, fuel, and fuel lines shall be subjected to an atmospheric pressure of 10.02 ± .05 psi simulating an altitude of 8,000 feet above sea level. The stove shall perform the standard cycle as specified in 4.6. The operational cycle shall be repeated 2 times. Verify that the stove operates at a simulated altitude of 8,000 feet.

4.6.44 Rain test. The stove shall be placed, unprotected, on the floor of the test chamber. The stove shall be disconnected from all power and fuel lines. It shall be subjected to rain at a rate of 4 in/hr for 10 minutes. Wind speed shall be 0 mph. The stove shall then be prepared for operation according to the manual. Prep time shall take no more than 5 minutes. The stove shall be operated for one standard cycle as specified in 4.6. Verify that the stove operates properly after being subjected to the above mentioned rain test.

4.6.45 Wind test. The standard test setup shall be used. The unit shall be subjected to a steady 10 mph wind. The stove shall perform a standard cycle as specified in 4.6. This procedure shall be repeated three times with the wind direction shifted 90 degrees each time. Verify that the stove operates properly after being subjected to the above mentioned wind test.

4.6.46 Vibration test. The stove, configured for transport according to the operators manual, shall be tested in accordance with the following procedure.

4.6.46.1 Test apparatus. A standard commercial package tester shall be used with fixturing as shown in figure 16. A

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vertical impact wall and sideboards as depicted shall be installed to contain the test items on the bed of the package tester. The fence opposite the vertical impact wall is not intended as an impact surface, but is used to restrain the test item from leaving the tester. The distance to this restraining fence should be sufficient to prevent constant impact. The sideboards shall be constructed such that the test item is constrained to a horizontal motion of 6 inches in a direction parallel to the axes of the shafts. The test item shall be initially positioned in the center of the space. The height of the test enclosure (sideboards, impact wall, and restraining wall) should be at least 2 inches higher than the height of the test item to prevent impacting of the test item on the top of the enclosure. The bed of the tester shall be covered with a steel plate.

4.6.46.2 Test level and duration. The basic movement of the bed of the test equipment shall be a 1 inch diameter orbital path at 5 Hz, such as can be obtained on a standard package tester operating in the synchronous mode. The stove shall be tested continuously for a minimum of 20 minutes. The test item shall be inspected to establish its pre-test condition and functionality. The test item shall then be placed on the bed of the package tester (see figure 16) as configured for field transport. The orientation of the item should represent its most likely transporting orientation. Any changes in orientation during the test shall not be corrected. The test apparatus shall be operated to provide the test level and duration specified. The test item shall be inspected for damage then it shall perform one standard operation test as specified in 4.6. Verify that the stove, as configured for field transport, operates properly after being subjected to the vibration test in the package tester.

4.6.47 Shock test. A stove, fully fueled, shall be raised 12 inches above a concrete surface and dropped 5 times to simulate a drop from the top burner position of the M59 range cabinet. The stove shall then be placed in the standard test setup as specified in 4.6. The stove shall then perform one standard cycle as specified in 4.6. Verify that the stove has no fuel leaking, and that it will operate properly after being subjected

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to the shock test.

4.6.48 Storage test. The burner units (stoves) shall operate after being subjected to the following conditions.

4.6.48.1 Low temperature storage test. The stove shall be subject to a temperature of -50 ± 2°F for a period of eight hours. The room temperature shall then be increased to normal ambient room temperature. After allowing the temperature of the stove to reach equilibrium, the stove shall perform one standard cycle as specified in 4.6. Verify that the stove does not fail to operate, or that permanent set or loss of resiliency of gaskets, discoloring, cracking, or bulging of any components has occurred.

4.6.48.2 High temperature storage test. The stove shall be subjected to a temperature of 160 ± 2°F for a period of eight hours. The room temperature shall then be decreased to normal ambient room temperature. After allowing the temperature of the stove to reach equilibrium, the stove shall perform one standard cycle as specified in 4.6. Verify that the stove does not fail to operate, or that permanent set or loss of resiliency of gaskets, discoloring, cracking, or bulging of any components has occurred.

4.6.49 Non-level operation test. This test may be conducted in conjunction with the reliability test in 4.6.50. The standard test setup as specified in 4.6 shall be used. The burner base shall be placed on a 4° slope. The stove shall perform a standard cycle as specified in 4.6. This procedure shall be repeated three times with the slope direction shifted 90 degrees each time for a total of 4 tests. Verify that the stove operates properly when tilted 4° on each of its four sides.

4.6.50 Reliability test. Four burners shall be operated a minimum of 300 hours each for a total of 1280 hours with not more than two Essential Function Failures (EFF) total for all four burners. An essential function shall be defined as the minimum operational tasks that the stove must perform to complete an operational cycle. For the purposes of this test an operational

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cycle shall be defined as one iteration of refueling the burner to a full condition, igniting the burner, operating it for a minimum of one hour, then manually shutting the stove off. Individual operational cycles may be any combination of high, medium, and low heat outputs. A failure shall be defined and any essential function failure requiring unit or depot unscheduled maintanance; any essential function failure requiring an operator unscheduled maintenance action that requires more that 30 minutes to diagnose, remove, repair/replace, and bring the stove back to an operational state following the procedures set forth in the operators manual; failure to light the stove within 3 atttempts within a 20 minute period; and Catastrophic and Critical (category I and II) safety hazards. Stoves shall be maintained and operated according to the instructions in the operators manual. Throughout the performance of the reliability test, verify that the stove meets the specified requirements.

4.6.51 Maintainability test. Maintenance data, including time required for each scheduled and unscheduled, maintenance or repair action shall be collected during performance of the reliability test (4.6.50). Verify that the stove is capable of having a maintenance ratio no greater than that required.

4.6.52 Cleaning test. This test shall be conducted in conjunction with the reliability test in 4.5.49. Verify that the stove can be adequately cleaned following the instructions in the manual. Verify that the unit can be returned to service after being cleaned according to the directions in the operators manual.

4.6.53 Nameplate verification. Verify that the name plate contains the required information.

4.6.54 Preventative Maintenance Checks and Services (PMCS) Verification. This test shall be conducted in conjunction with the reliability test in 4.6.50. Verify that PMCS as described in the operators manual can be performed within the specified time 95% of the time.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The stove is intended for use as the primary heat for military field feeding equipment. It provides heat for grilling, roasting, frying, boiling, sanitation, and baking. It can be expected that food, water, and grease will be spilled on the stove. Cleaning the stove will be required using commonly available cleaning solvents and scouring pads. It will be used in environmental conditions hot through basic. The unit will be ignited and operated inside tents, soft sided trailers (MKT), and soft sided expandable shelters. The unit will be exposed to, but will not be operated in, the rain. The stove may be transported on ground vehicles as unrestrained cargo. All maintenance required will be performed by the operator without any special tools. As much as practical, the operator will perform all repairs and replacement of components using no special tools.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. When a first article is required (see 3.1, 4.2, 6.3, and 6.6).
c. Sampling procedures for first article testing and conformance testing (see 4.2 and 4.3).
d. Sampling procedures for acceptance testing (see 4.4)
e. Packaging requirements (see 5.1).
f. Government furnished equipment source and delivery (see 6.3).

6.3 Government Furnished Equipment (GFE). The following GFE will be required for interface and compatibility purposes for first article and conformance inspection.

Item	Identifying Number	Quantity
Heater tank assembly	NSN 7360-01-248-6041	1
Mobile Kitchen Trailer (MKT)	NSN 7360-01-155-6020	1
M59 range cabinet	NSN 7360-00-028-2153	2
Field sanitation center	NSN 7360-01-277-2558	1
Fuel can	NSN 7240-01-337-5269	1
Spout, can, gasoline	NSN 7240-00-177-6154	1
Suction feed adapter	NSN 7240-21-912-7125	1
Fuel line		1
Power cable (25 ft)		1
Power cable (2.5 ft)		2
Cold weather glove	NSN 8415-01-310-7338.	1 pair

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6.4 Subject term (keyword) listing.

Cooking
Fuel
Heat
JP-8
Stove

6.5 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided the material meets or exceeds all specified requirements and promotes economically advantageous life cycle costs.

6.6 First article. When requiring a first article inspection, the contracting officer should provide specific guidance to offerors whether the first article is a first article sample, a first production item, or a number of items to be tested. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval or first article test results and disposition of first articles. Pre-solicitation documents should provide government waiver rights for samples for first article inspection to bidders offering a previously-acquired or tested product. Bidders offering such products, who wish to rely on such production or testing, must furnish evidence with the bid, that prior government approval is appropriate for the pending contract.

6.7 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include contractor past performance, government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the unitofmity of measuring processes and techniques.

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Custodians:

Army - GL
Air Force - 99

Review Activities:

Army - AV,MD1,QM1
Air Force - 35,84
DLA - IS

Preparing activity:

Army - GL
(Project 7310-0881)

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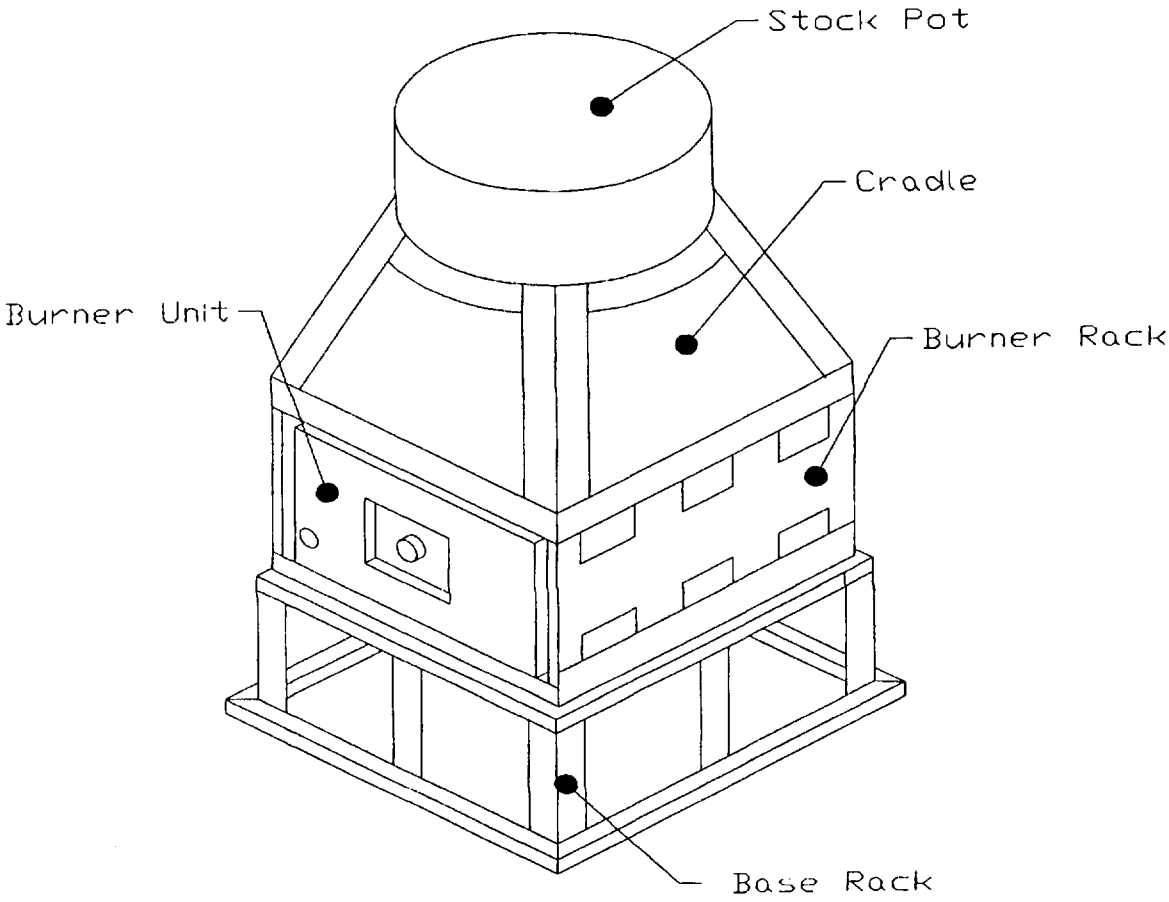


FIGURE 1. Standard test setup.

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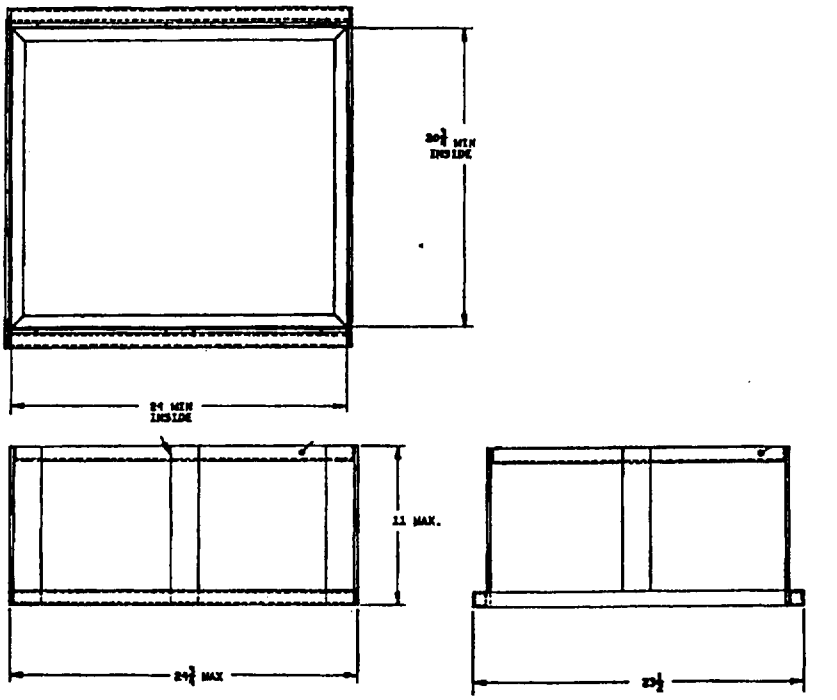


FIGURE 2. Base rack assembly.

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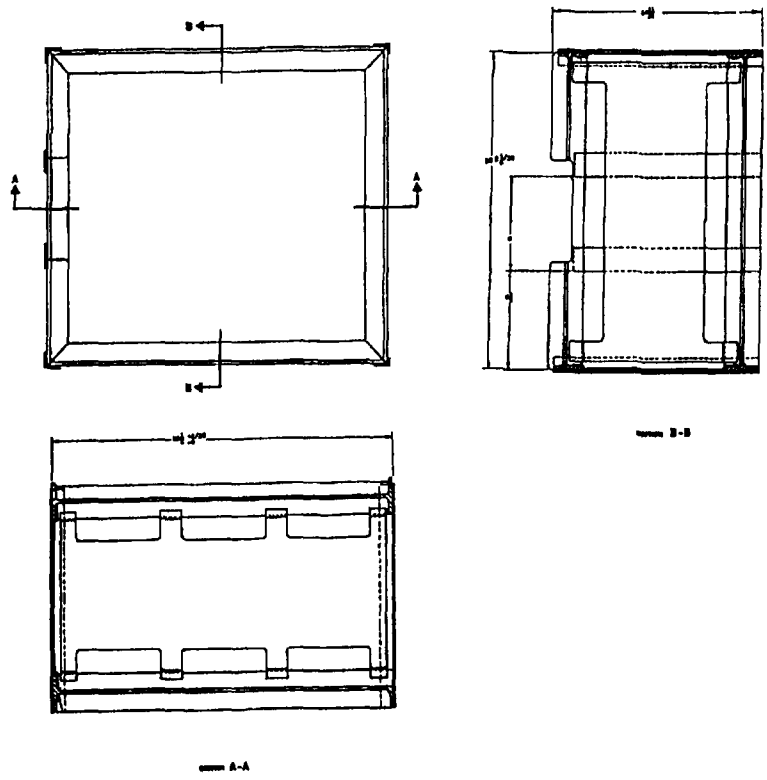


FIGURE 3. Burner rack assembly.

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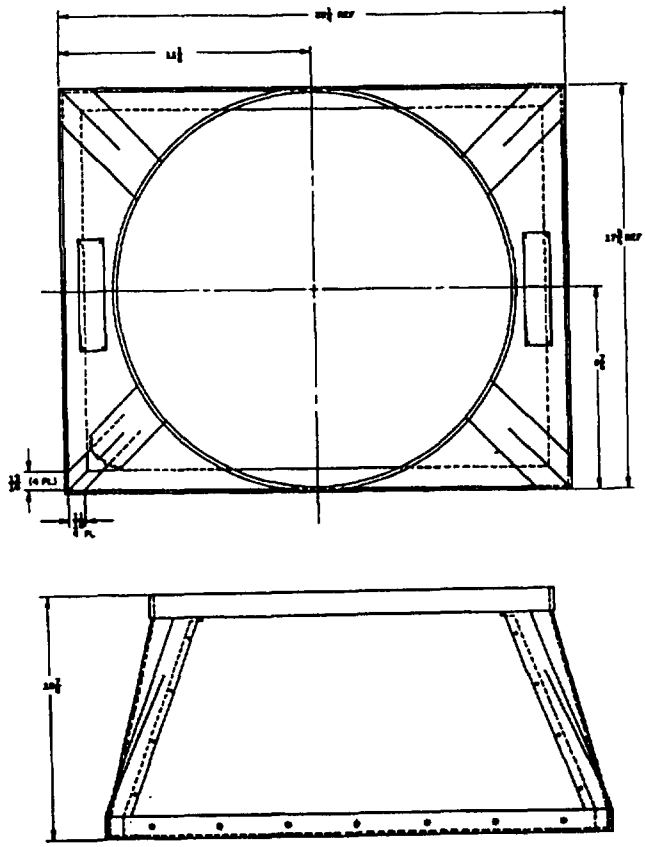


FIGURE 4. Cook pot cradle.

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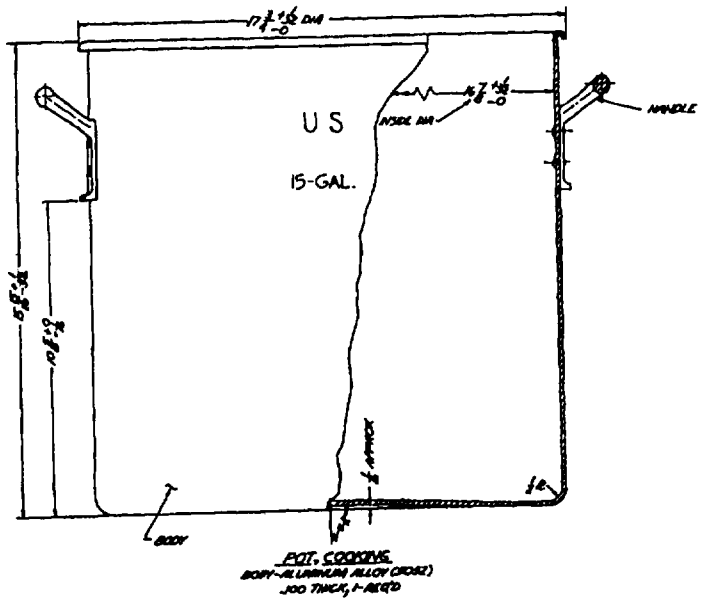


FIGURE 5. Pot, cooking, 15 gallon.

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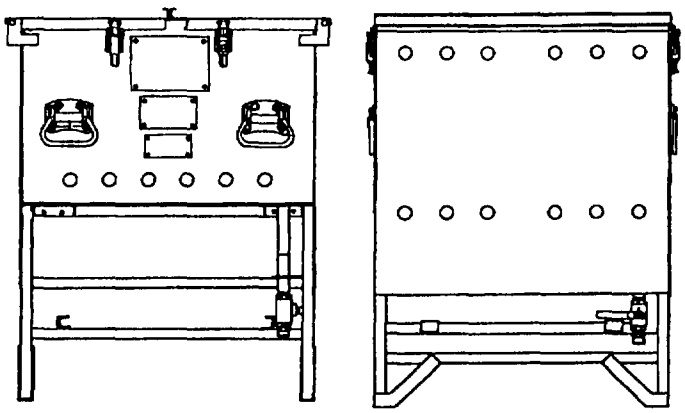


FIGURE 6. Tray pack heater tank assembly.

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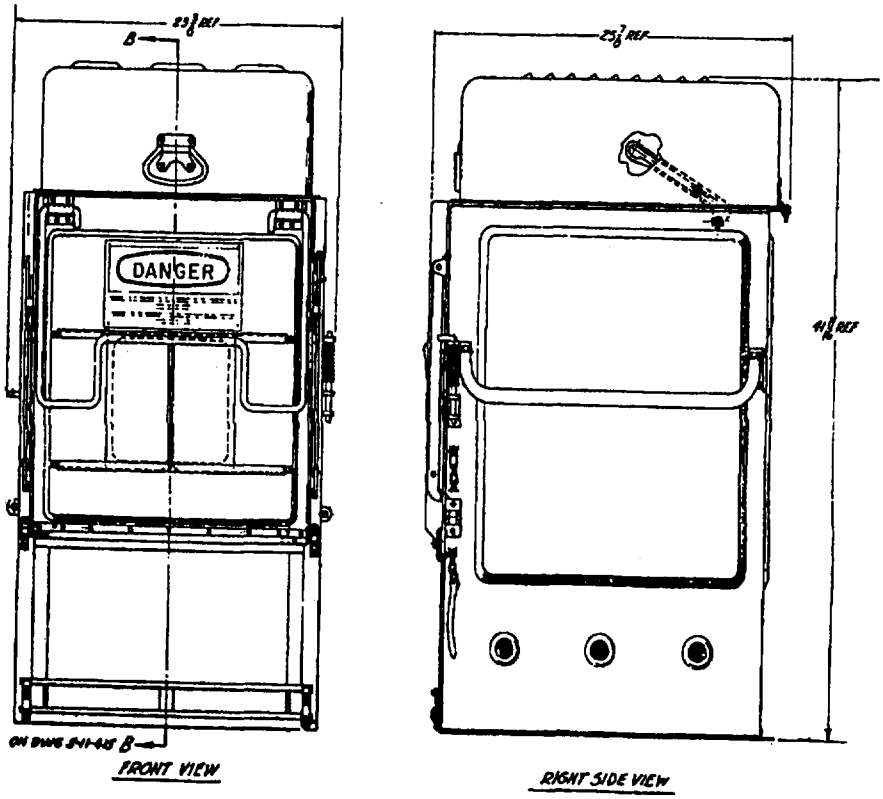


FIGURE 7. M59 field range outfit.

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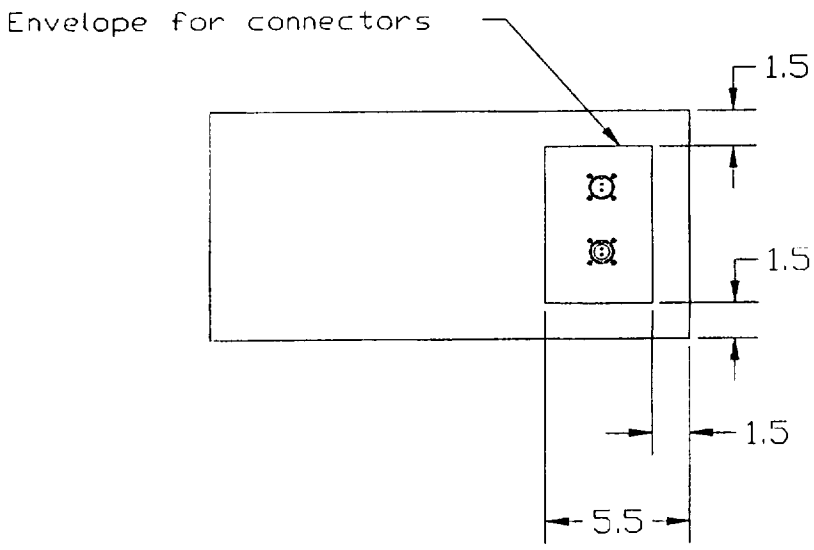


FIGURE 8. Location of electrical interface connectors.

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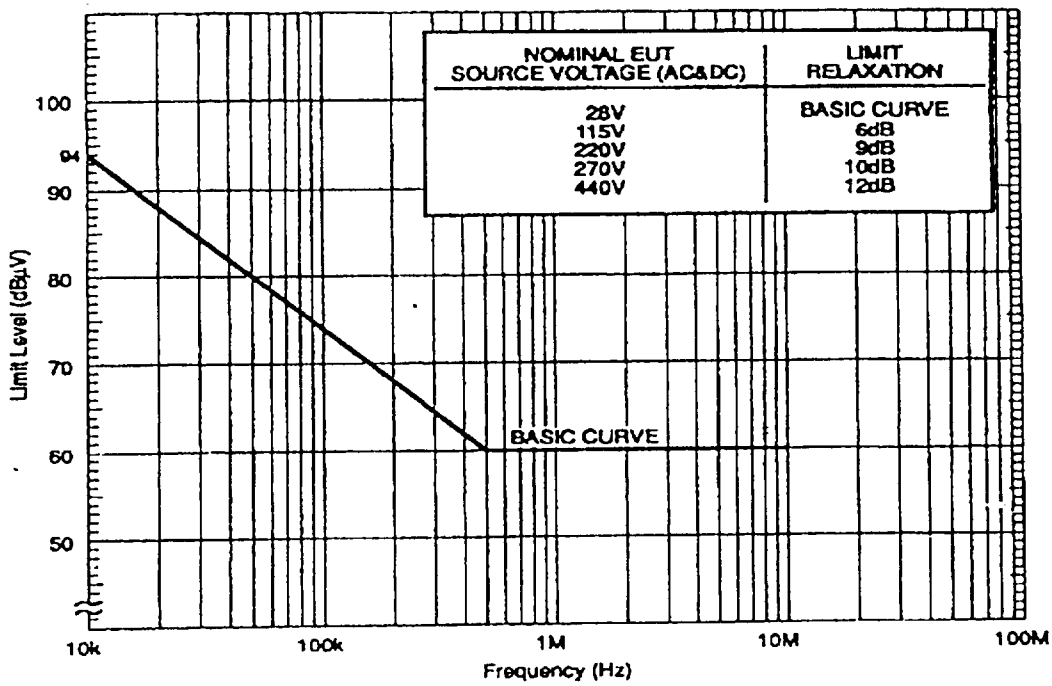


FIGURE 9. Conducted Emissions Limits for Stove.

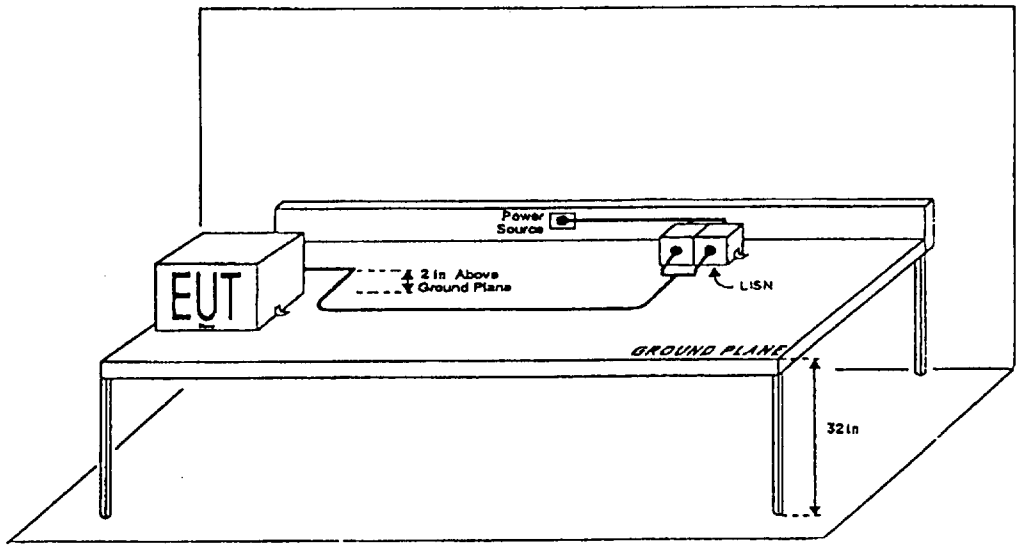


FIGURE 10. General test setup for conducted emissions.

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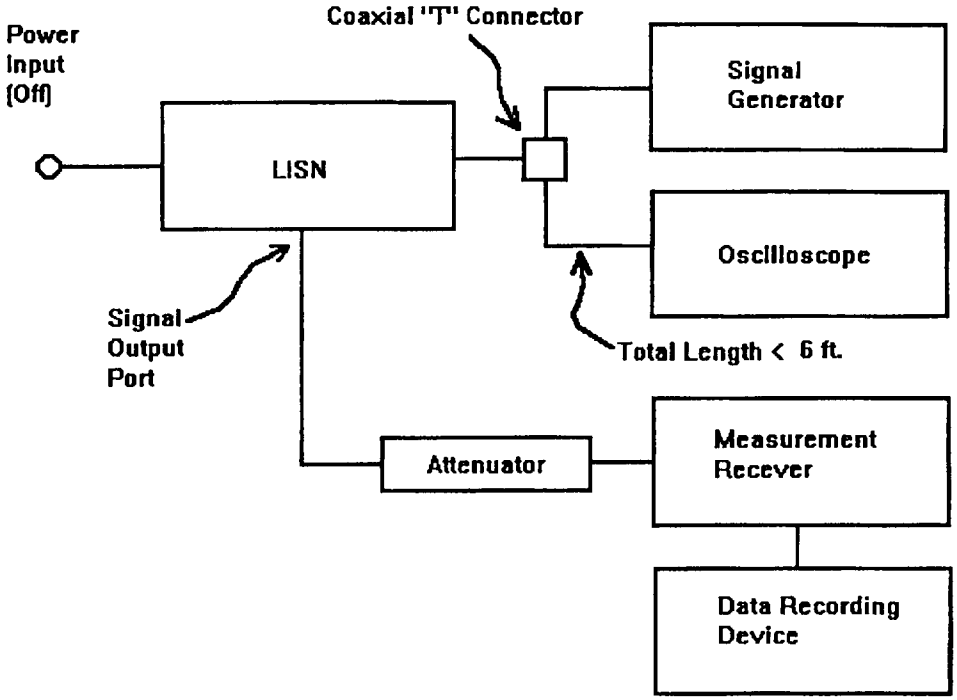


FIGURE 11. Measurement system check setup for conducted emissions.

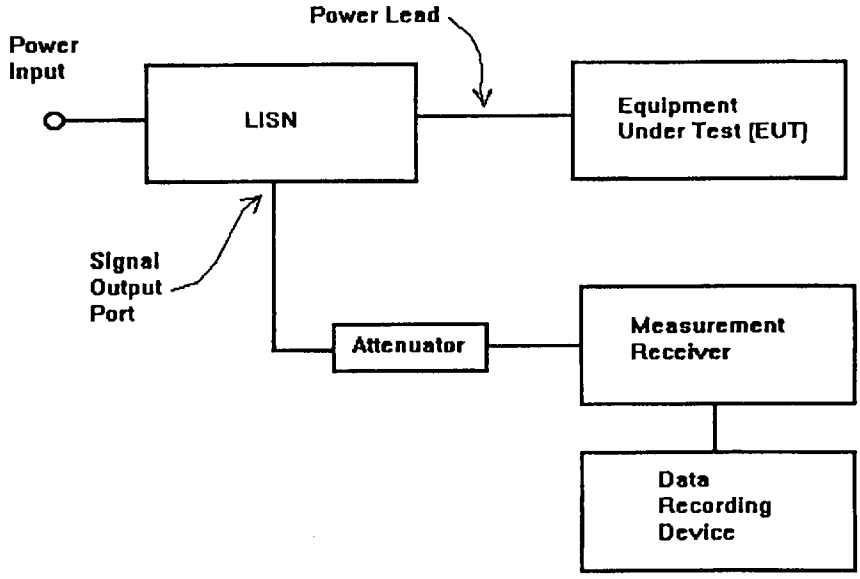


FIGURE 12. Measurement setup for conducted emissions.

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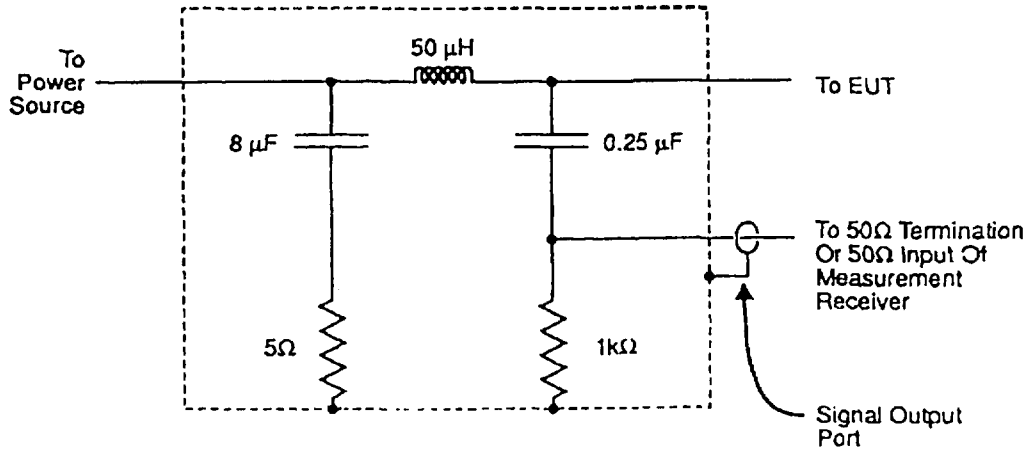


FIGURE 13. LISN schematic.

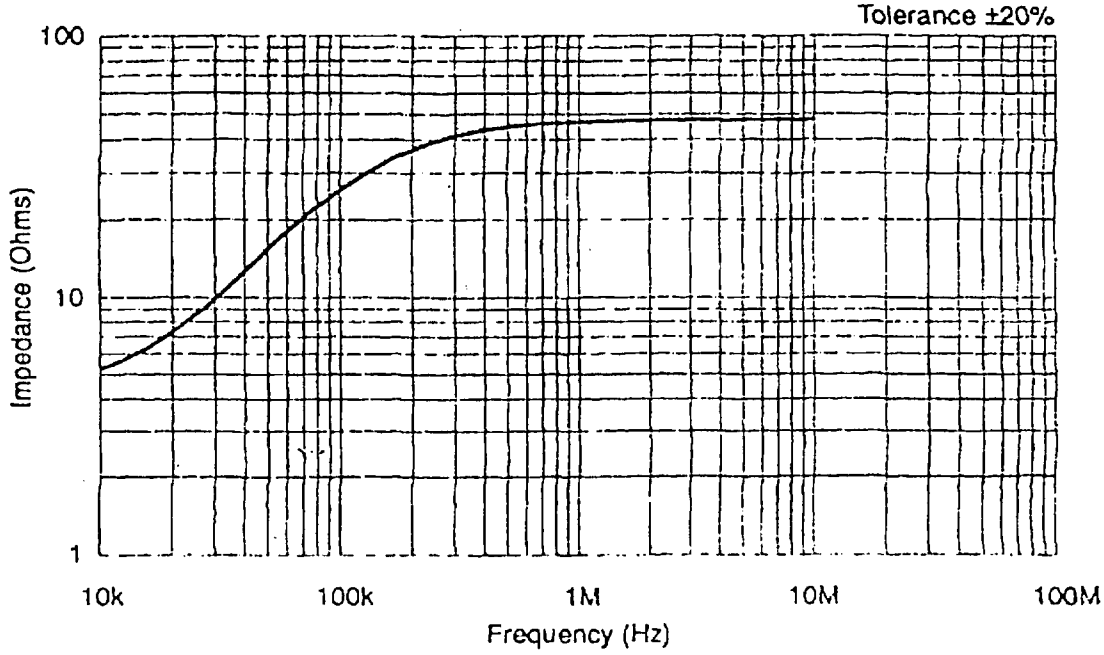


FIGURE 14. LISN Impedance.

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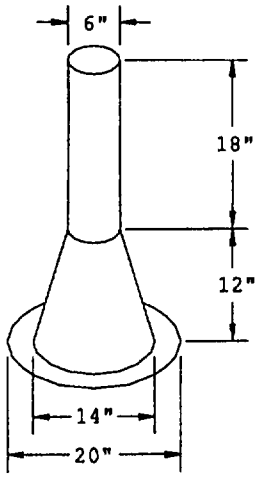


FIGURE 15. Combustion product collector.

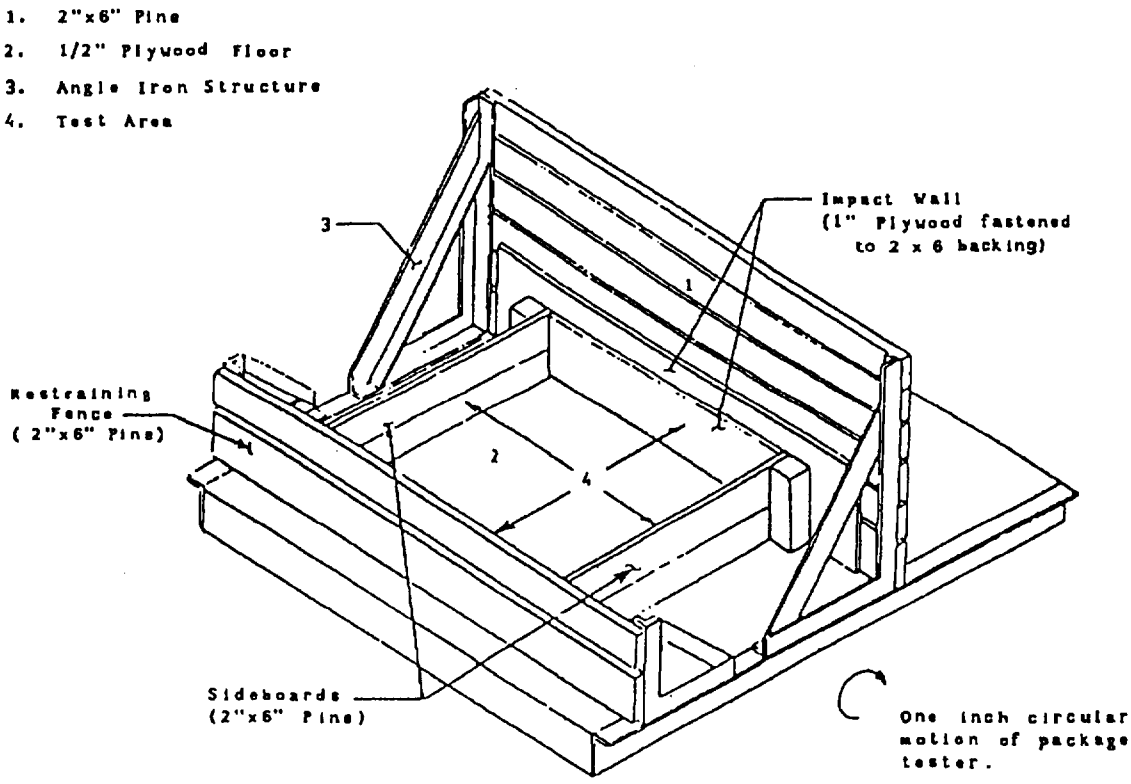


FIGURE 16. Typical package tester.

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